

# Machine Learning Experiment no 4 (Linear Regresssion)

## Libraries

```
In [1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

%matplotlib inline
```

## Load the dataset

```
In [2]: HouseDF = pd.read_csv('USA_Housing.csv')

In [3]: HouseDF.head()
```

Out[3]:

|   | Avg. Area<br>Income | Avg.<br>Area<br>House<br>Age | Avg.<br>Area<br>Number<br>of<br>Rooms | Avg. Area<br>Number<br>of<br>Bedrooms | Area<br>Population | Price        |                               |
|---|---------------------|------------------------------|---------------------------------------|---------------------------------------|--------------------|--------------|-------------------------------|
| 0 | 79545.458574        | 5.682861                     | 7.009188                              | 4.09                                  | 23086.800503       | 1.059034e+06 | 208 Michael F<br>674\nLaura   |
| 1 | 79248.642455        | 6.002900                     | 6.730821                              | 3.09                                  | 40173.072174       | 1.505891e+06 | 188 Johns<br>Suite 0<br>Kathl |
| 2 | 61287.067179        | 5.865890                     | 8.512727                              | 5.13                                  | 36882.159400       | 1.058988e+06 | 9127<br>Stravenue\nDa<br>W    |
| 3 | 63345.240046        | 7.188236                     | 5.586729                              | 3.26                                  | 34310.242831       | 1.260617e+06 | USS Barnettv                  |
| 4 | 59982.197226        | 5.040555                     | 7.839388                              | 4.23                                  | 26354.109472       | 6.309435e+05 | USNS Raymo<br>,               |

```
In [4]: HouseDF.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Avg. Area Income                      5000 non-null   float64
1   Avg. Area House Age                  5000 non-null   float64
2   Avg. Area Number of Rooms            5000 non-null   float64
3   Avg. Area Number of Bedrooms         5000 non-null   float64
4   Area Population                      5000 non-null   float64
5   Price                                5000 non-null   float64
6   Address                              5000 non-null   object
dtypes: float64(6), object(1)
memory usage: 273.6+ KB
```

In [5]: `HouseDF.describe()`

Out[5]:

|              | Avg. Area<br>Income | Avg. Area<br>House Age | Avg. Area<br>Number of<br>Rooms | Avg. Area<br>Number of<br>Bedrooms | Area<br>Population | Price        |
|--------------|---------------------|------------------------|---------------------------------|------------------------------------|--------------------|--------------|
| <b>count</b> | 5000.000000         | 5000.000000            | 5000.000000                     | 5000.000000                        | 5000.000000        | 5.000000e+03 |
| <b>mean</b>  | 68583.108984        | 5.977222               | 6.987792                        | 3.981330                           | 36163.516039       | 1.232073e+06 |
| <b>std</b>   | 10657.991214        | 0.991456               | 1.005833                        | 1.234137                           | 9925.650114        | 3.531176e+05 |
| <b>min</b>   | 17796.631190        | 2.644304               | 3.236194                        | 2.000000                           | 172.610686         | 1.593866e+04 |
| <b>25%</b>   | 61480.562388        | 5.322283               | 6.299250                        | 3.140000                           | 29403.928702       | 9.975771e+05 |
| <b>50%</b>   | 68804.286404        | 5.970429               | 7.002902                        | 4.050000                           | 36199.406689       | 1.232669e+06 |
| <b>75%</b>   | 75783.338666        | 6.650808               | 7.665871                        | 4.490000                           | 42861.290769       | 1.471210e+06 |
| <b>max</b>   | 107701.748378       | 9.519088               | 10.759588                       | 6.500000                           | 69621.713378       | 2.469066e+06 |

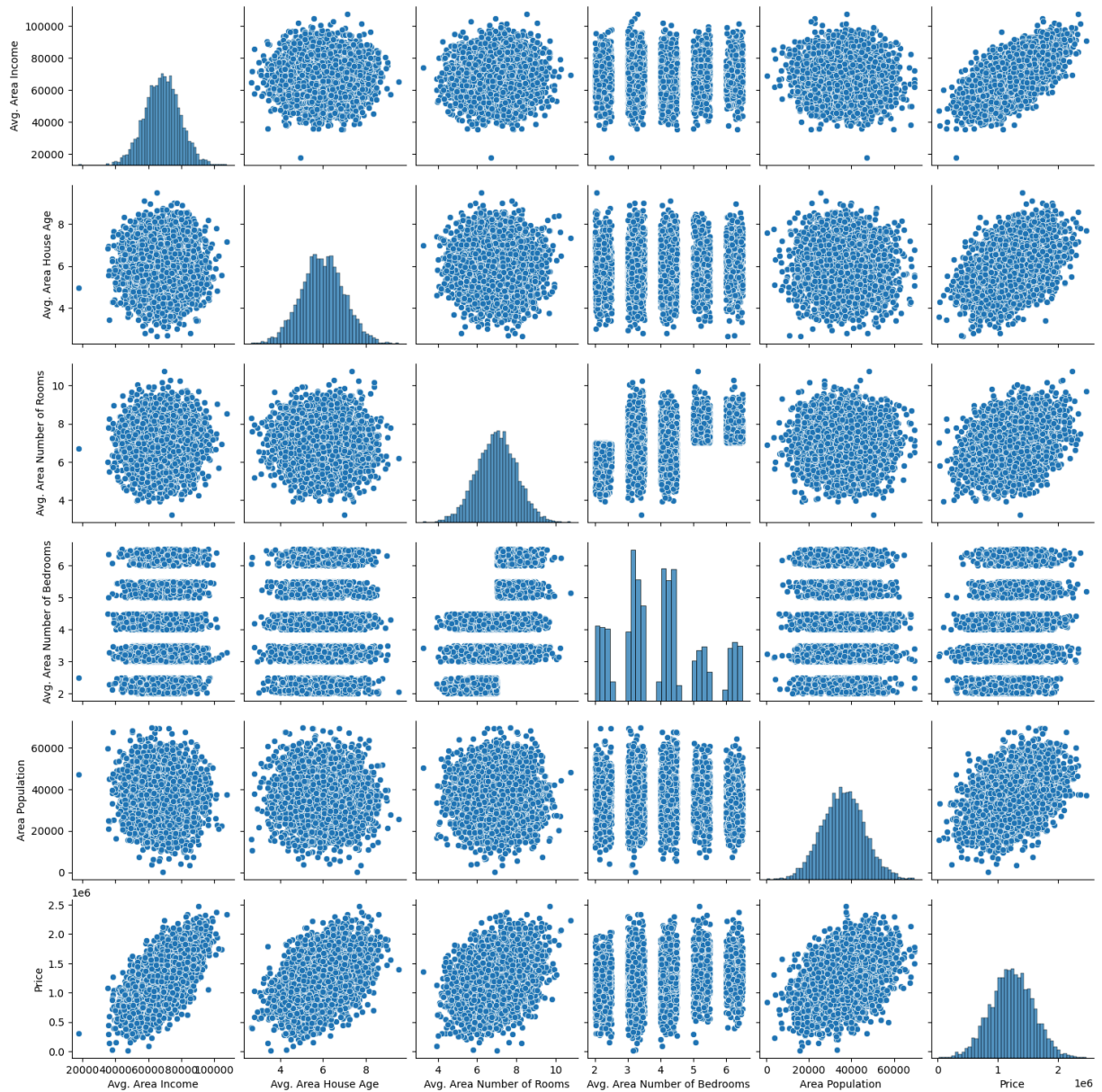
In [8]: `HouseDF.columns`

Out[8]: `Index(['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area Number of Rooms', 'Avg. Area Number of Bedrooms', 'Area Population', 'Price', 'Address'], dtype='object')`

## EDA

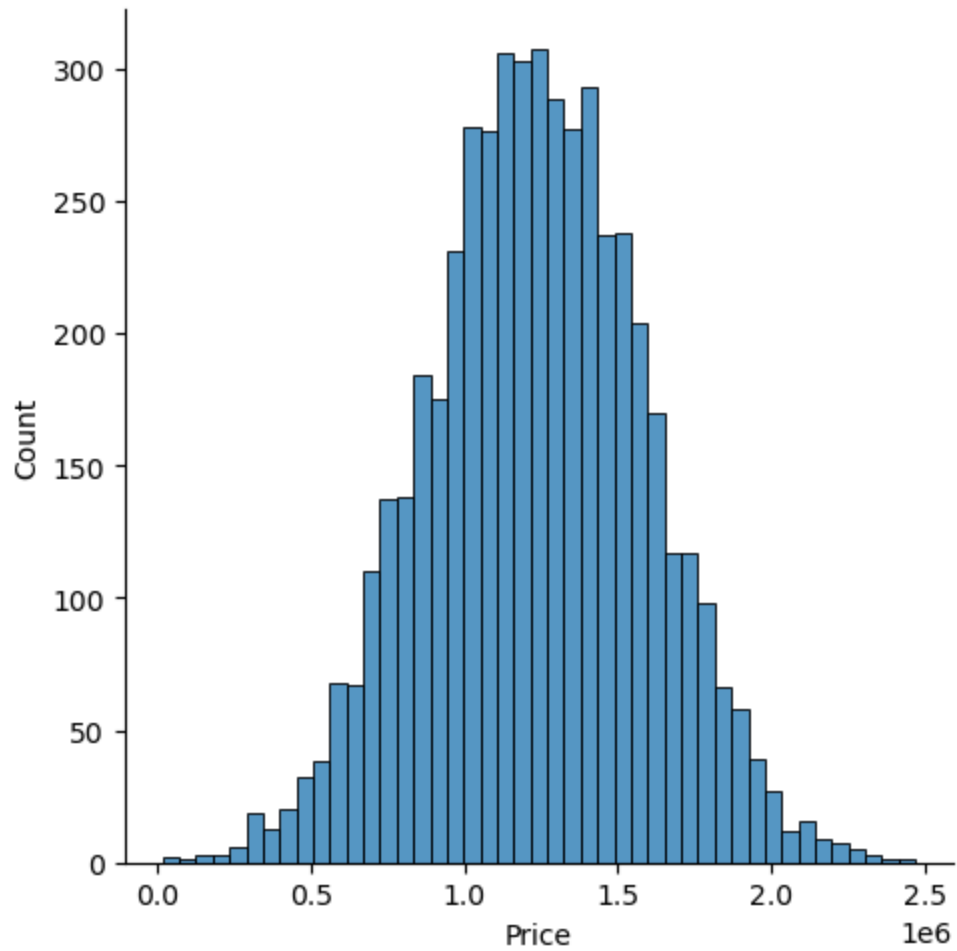
In [9]: `sns.pairplot(HouseDF)`

Out[9]: `<seaborn.axisgrid.PairGrid at 0x1f5abae8440>`



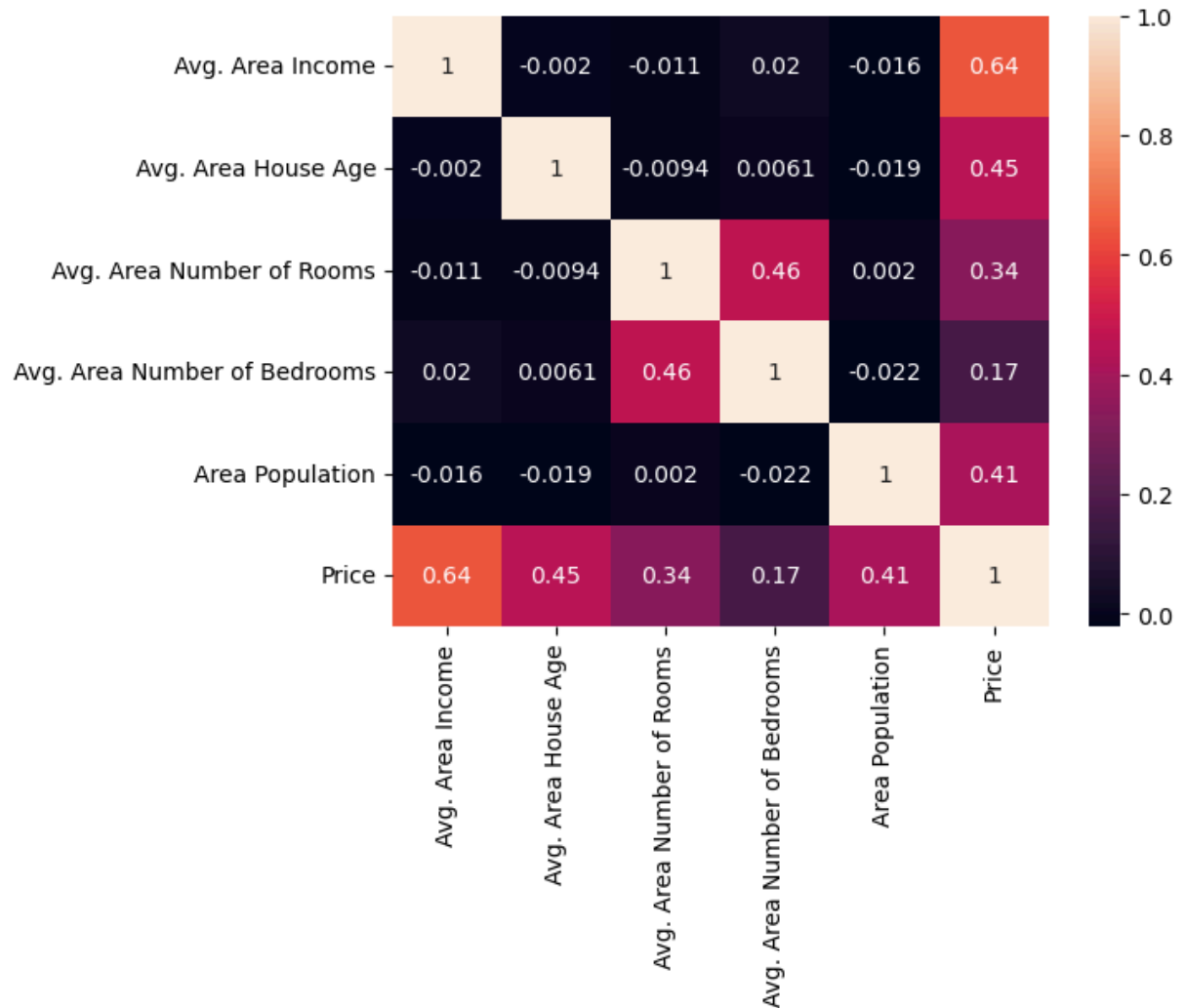
```
In [10]: sns.displot(HouseDF['Price'])
```

```
Out[10]: <seaborn.axisgrid.FacetGrid at 0x1f5ad5ac050>
```



```
In [13]: sns.heatmap(HouseDF.corr(numeric_only=True), annot=True)
```

```
Out[13]: <Axes: >
```



## Model Training

```
In [14]: X = HouseDF[['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area Number of Rooms',
                    'Avg. Area Number of Bedrooms', 'Area Population']]

y = HouseDF['Price']
```

```
In [15]: from sklearn.model_selection import train_test_split
```

```
In [16]: X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.4, random_state=42)
```

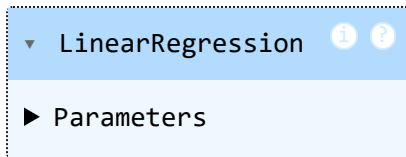
## Creating and training model

```
In [17]: from sklearn.linear_model import LinearRegression
```

```
In [18]: lm = LinearRegression()
```

```
In [19]: lm.fit(X_train, y_train)
```

Out[19]:

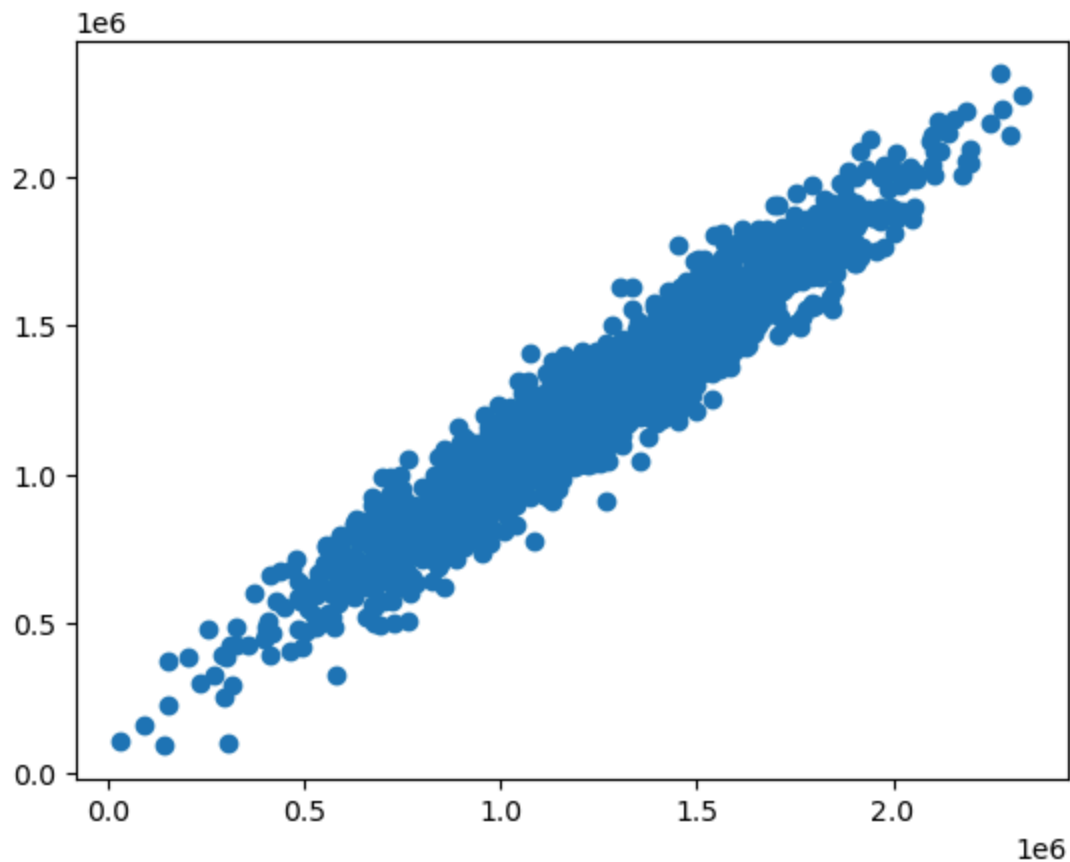
In [20]: `print(lm.intercept_)``-2640159.79685191`In [21]: `coeff_df = pd.DataFrame(lm.coef_,X.columns,columns=['Coefficient'])`  
`coeff_df`

Out[21]:

|                              | Coefficient   |
|------------------------------|---------------|
| Avg. Area Income             | 21.528276     |
| Avg. Area House Age          | 164883.282027 |
| Avg. Area Number of Rooms    | 122368.678027 |
| Avg. Area Number of Bedrooms | 2233.801864   |
| Area Population              | 15.150420     |

## Prediction from model

In [22]: `predictions = lm.predict(X_test)`In [23]: `plt.scatter(y_test,predictions)`Out[23]: `<matplotlib.collections.PathCollection at 0x1f5b0a22fd0>`



```
In [26]: sns.distplot((y_test-predictions),bins=50);
```

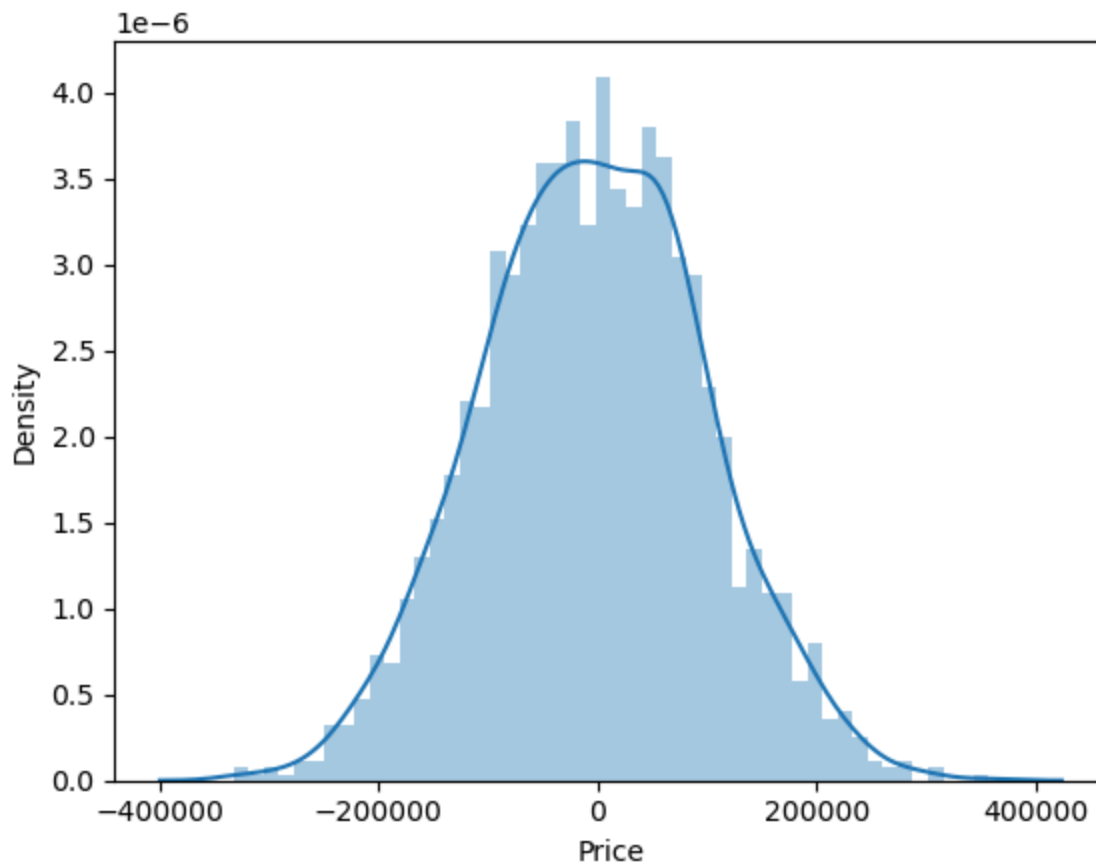
C:\Users\HP\AppData\Local\Temp\ipykernel\_16072\1326397652.py:1: UserWarning:

``distplot` is a deprecated function and will be removed in seaborn v0.14.0.`

Please adapt your code to use either ``displot`` (a figure-level function with similar flexibility) or ``histplot`` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot((y_test-predictions),bins=50);
```



## Evaluation Metrics

```
In [27]: from sklearn import metrics
```

```
In [28]: print('MAE:', metrics.mean_absolute_error(y_test, predictions))  
         print('MSE:', metrics.mean_squared_error(y_test, predictions))  
         print('RMSE:', np.sqrt(metrics.mean_squared_error(y_test, predictions)))
```

```
MAE: 82288.22251914954  
MSE: 10460958907.209503  
RMSE: 102278.82922291153
```